

HDOC Day-31

Question: Write a C program to perform menu-driven operations on a given 2D array: (1) find the sum of all elements, (2) find the sum of each row and store it in an array, (3) find the transpose of the matrix, and (4) exit.

C Program

```
#include <stdio.h>

int main()
{
    int a[10][10], transpose[10][10];
    int rowSum[10];
    int rows, cols, i, j, choice, sum;

    printf("Enter number of rows and columns: ");
    scanf("%d %d", &rows, &cols);

    printf("Enter matrix elements:\n");
    for(i = 0; i < rows; i++)
    {
        for(j = 0; j < cols; j++)
            scanf("%d", &a[i][j]);
    }

    do
    {
        printf("\n--- MENU ---\n");
        printf("1. Sum of all elements\n");
        printf("2. Sum of each row\n");
        printf("3. Transpose of matrix\n");
        printf("4. Exit\n");
        printf("Enter your choice: ");
        scanf("%d", &choice);

        switch(choice)
        {
            case 1:
                sum = 0;
                for(i = 0; i < rows; i++)
                    for(j = 0; j < cols; j++)
                        sum = sum + a[i][j];

                printf("Sum of all elements = %d\n", sum);
                break;

            case 2:
                for(i = 0; i < rows; i++)
                {
                    rowSum[i] = 0;
                    for(j = 0; j < cols; j++)
                        rowSum[i] = rowSum[i] + a[i][j];
                }

                printf("Sum of each row:\n");
                for(i = 0; i < rows; i++)
                    printf("Row %d = %d\n", i + 1, rowSum[i]);
                break;

            case 3:
                for(i = 0; i < rows; i++)
                    for(j = 0; j < cols; j++)
                        transpose[j][i] = a[i][j];

                printf("Transpose of matrix:\n");
                for(i = 0; i < cols; i++)
                {
                    for(j = 0; j < rows; j++)
                        printf("%d\t", transpose[i][j]);
                }
            }
        }
    } while(choice != 4);
}
```

```

        printf("\n");
    }
    break;

case 4:
    printf("Program terminated.\n");
    break;

default:
    printf("Invalid choice! Try again.\n");
}
} while(choice != 4);

return 0;
}

```

Sample Output

```

Enter number of rows and columns: 2 3
Enter matrix elements:
1 2 3
4 5 6

```

```

--- MENU ---
1. Sum of all elements
2. Sum of each row
3. Transpose of matrix
4. Exit
Enter your choice: 1
Sum of all elements = 21

```

```

Enter your choice: 2
Sum of each row:
Row 1 = 6
Row 2 = 15

```

```

Enter your choice: 3
Transpose of matrix:
1 4
2 5
3 6

```

```

Enter your choice: 4
Program terminated.

```

Result: The program successfully performs the selected operations on a 2D matrix using a menu-driven approach.